

PATENT LANDSCAPE REPORT: LOW MOLECULAR WEIGHT NBR AND RELATED LOW-VISCOSITY POLYMER SYSTEMS

1. ABSTRACT

This report synthesizes a patent set returned for the target concept of low molecular weight NBR, but the retrieved primary patents are largely adjacent rather than direct disclosures of low-MW nitrile-butadiene rubber synthesis. Across the manifest, only US12146207B2 explicitly includes NBR as a rubber component in degradable wellbore diverters/seals, while EP2389295B1 concerns functionalized diene rubbers in silica-filled tire tread compounds, EP2921509B1 discloses liquid ethylene/alpha-olefin copolymers prepared by solution polymerization with bridged metallocene catalysts, US10465017B2 describes essentially adiabatic bulk polymerization of silicone-containing copolymers using thermal initiators and chain-transfer agents, and US10161556B2 uses HNBR as a high-temperature insulation material in subsea pipelines. None of the retrieved patents provides a direct, dedicated route to low molecular weight NBR via controlled polymerization, depolymerization, or chain-transfer engineering. However, the set is still informative because it highlights the process levers most relevant to low-MW elastomer design: solution polymerization, catalyst/initiator selection, chain-transfer control, thermal management, and formulation-driven viscosity or stiffness tuning. EP2921509B1 is the closest process analogue in terms of deliberately targeting liquid, low-Mn polymer products with narrow molecular-weight distribution, while US10465017B2 demonstrates staged adiabatic polymerization and explicit chain-transfer use to manage conversion and polymer properties. EP2389295B1 and US10161556B2 show how diene-rubber performance is often governed more by compounding and end-use requirements than by direct low-MW synthesis. US12146207B2 confirms that NBR can be used as a degradable elastomeric component in temporary sealing systems, but it does not disclose NBR molecular-weight control. Overall, the landscape suggests that low molecular weight NBR development is most likely to rely on solution polymerization with controlled chain transfer, carefully selected initiator systems, and tight temperature control, rather than on the formulations disclosed in these retrieved patents.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2389295B1
- Patent Title: Rubber mixtures containing silane and having possibly functionalized diene ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses rubber mixtures for tire treads based on a functionalized diene rubber, optionally combined with an SBR microgel and silane coupling agent. The functionalized diene rubber is described as being prepared by solution polymerization of 1,3-butadiene and styrene followed by introduction of functional groups such as hydroxy and/or carboxy groups. - 1,5-L kneader mixing time: 6 minutes — Example 2 - Temperature rise during first mixing stage: 70 to 154°C — Example 2 - Hold temperature during first mixing stage: 150°C for 3 minutes — Example 2 - Second mixing stage: 3 minutes at 150°C — Example 2 - Roller mixing temperature: 40-60°C — Example 2 - Mooney viscosity ML 1+1 (100°C): 61.3, 67.5, 69.3, 80.0 ME — Table 2 - Mooney viscosity ML 1+4 (100°C): 55.6, 61.2, 63.2, 73.1 ME — Table 2 - Mooney relaxation /10 sec: 21.0, 22.0, 22.7, 25.3% — Table 2 - Mooney relaxation /30 sec: 14.6, 15.5, 16.3, 18.8% — Table 2 - Vulcanization temperature: 160°C — Table 3 - Press vulcanization time: 20 minutes at 160°C — Table 4 - Shore A hardness at 23°C: 59.1, 59.7, 58.7, 61.6 — Table 4 - Shore A hardness at 70°C: 57.9, 57.3, 56.5, 59.0 — Table 4

Relevant Experimental Evidence - Example 1: baseline rubber blend comparison was introduced, but no structured synthesis parameters were extracted. - Example 2: detailed compounding recipe and two-stage mixing protocol were disclosed for functionalized diene rubber/silica/silane blends. - Example 3: inventive examples were identified in the formulation table, but no additional process details were extracted. - Example 4: Mooney relaxation and vulcanization behavior were quantified for the prepared blends. - Example 5: vulcanizate hardness and related properties were reported after press curing.

Technical Relevance Although not an NBR synthesis patent, it is relevant because it shows how diene-rubber properties are tuned through functionalization, mixing history, and silica/silane formulation. The Mooney and cure data are useful analogues for understanding viscosity and processability targets in low-MW elastomer systems.

Patent 2

- Patent Number: EP2921509B1
- Patent Title: Method for preparing ethylene and alpha-olefin copolymers
- Assignee: Not disclosed
- Jurisdiction: EP

- Publication Year: 2025
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution polymerization of ethylene and C3-C20 alpha-olefins to produce liquid random copolymers using a bridged metallocene catalyst system and an ion-pair-forming cocatalyst. The process is explicitly designed to obtain low-Mn, low-viscosity liquid copolymers with controlled molecular-weight distribution. - Ethylene feed: 260 g/hr — Comparative Example 3 - Propylene feed: 457 g/hr — Comparative Example 3 - Hydrogen feed: 0.65 g/hr — Comparative Example 3 - Hexane feed: 30 g/hr — Comparative Example 3 - Catalyst loading: 2.3 mg/hr dimethylsilylbis(indenyl)zirconium dichloride — Comparative Example 3 - Catalyst loading: 3.8 mg/hr diphenylmethylene(3-n-butyl-cyclopentadienyl)(2,7-di-tert-butylfluorenyl)zirconium dichloride — Comparative Example 3 - Promoter loading: 4.2 mL/hr methylaluminoxane — Comparative Example 3 - Promoter loading: 5.6 g/hr dimethylanilinium tetra(pentafluorophenyl)boron — Comparative Example 3 - Reaction temperature: 100°C — Comparative Example 3 - Pressure: 16 psig — Comparative Example 3 - Ethylene amount: 45.08 mol% — Example 1 - Mn: 2080 — Example 1 - Mw/Mn: 1.82 — Example 1 - Kinematic viscosity at 100°C: 140 cSt — Example 1 - Kinematic viscosity at 40°C: 1900 cSt — Example 1 - Pour point: -30°C — Example 1 - Flash point: 247°C — Example 1 - Bromine number: < 0.1 — Example 1 - Appearance: Clear — Example 1 - Ethylene amount: 45.07 mol% — Example 2 - Mn: 3736 — Example 2 - Mw/Mn: 1.81 — Example 2 - Kinematic viscosity at 100°C: 700 cSt — Example 2 - Kinematic viscosity at 40°C: 7300 cSt — Example 2 - Pour point: -15°C — Example 2 - Flash point: 280°C — Example 2 - Bromine number: < 0.1 — Example 2 - Appearance: Clear — Example 2 - Ethylene amount: 45.17 mol% — Example 3 - Mn: 5385 — Example 3 - Mw/Mn: 1.88 — Example 3 - Kinematic viscosity at 100°C: 2300 cSt — Example 3 - Kinematic viscosity at 40°C: 44100 cSt — Example 3 - Pour point: 0.0°C — Example 3 - Flash point: 290°C — Example 3 - Bromine number: < 0.1 — Example 3 - Appearance: Clear — Example 3 - Ethylene amount: 45.10 mol% — Reference Example 4 - Mn: 2135 — Reference Example 4 - Mw/Mn: 1.82 — Reference Example 4 - Kinematic viscosity at 100°C: 145 cSt — Reference Example 4 - Kinematic viscosity at 40°C: 2000 cSt — Reference Example 4 - Pour point: -29.5°C — Reference Example 4 - Flash point: 245°C — Reference Example 4 - Bromine number: 0.5 — Reference Example 4 - Appearance: Haze — Reference Example 4 - Ethylene amount: 45.01 mol% — Reference Example 5 - Mn: 3694 — Reference Example 5 - Mw/Mn: 1.83 — Reference Example 5 - Kinematic viscosity at 100°C: 750 cSt — Reference Example 5 - Kinematic viscosity at 40°C: 8100 cSt — Reference Example 5 - Pour point: -12.5°C — Reference Example 5 - Flash point: 284°C — Reference Example 5 - Bromine number: 0.4 — Reference Example 5 - Appearance: Haze — Reference Example 5 - Ethylene amount: 45.89 mol% — Reference Example 6 - Mn: 5985 — Reference Example 6 - Mw/Mn: 1.89 — Reference Example 6 - Kinematic viscosity at 100°C: 2450 cSt — Reference Example 6 - Kinematic viscosity at 40°C: 49500 cSt — Reference Example 6 - Pour point: 0.0°C — Reference Example 6 - Flash point: 301°C — Reference Example 6 - Bromine number: 0.3 — Reference Example 6 - Appearance: Haze — Reference Example 6

Relevant Experimental Evidence - Example 1: produced a liquid copolymer with Mn 2080, Mw/Mn 1.82, and bromine number <0.1. - Example 2: increased Mn to 3736 while maintaining clear appearance and low bromine number. - Example 3: further increased Mn to 5385 and retained clear appearance with low bromine number. - Comparative Example 3: catalyst and promoter conditions were listed for the polymerization system. - Reference Examples 4-6: higher bromine numbers and haze were associated with less favorable product quality.

Technical Relevance This is the closest process analogue in the set for low molecular weight polymer manufacture because it explicitly targets liquid, low-Mn products with controlled viscosity. The metallocene solution-polymerization framework is highly relevant as a conceptual model for low-MW NBR synthesis via controlled molecular-weight build.

Patent 3

- Patent Number: US10465017B2
- Patent Title: Bulk polymerization of silicone-containing copolymers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses essentially adiabatic bulk polymerization of silicone-containing copolymers using free-radical monomers, thermal initiators, and a chain transfer agent separate from the mercapto-functional silicone. The process is staged: a first polymerization is carried out, then additional initiator is added and a second adiabatic polymerization step is performed to complete the polymer. - IOA: 85 parts by weight — Example 1 - AA: 10 parts by weight — Example 1 - SiMac: 5 parts by weight — Example 1 - Initiator-1: 0.004 parts by weight — Example 1 - IOTG: 0.045 parts by weight — Example 1 - ABP: 0.3125 parts by weight — Example 1 - IOA: 80 parts by weight — Example 2 — Table 1 - AA: 10 parts by weight — Example 2 — Table 1 - SiMac: 10 parts by weight — Example 2 — Table 1 - Initiator-1: 0.004 parts by weight — Example 2 — Table 1 - IOTG: 0.06 parts by weight — Example 2 — Table 1 - ABP: 0.3125 parts by weight — Example 2 — Table 1 - IOA: 80 parts by weight — Example 3 — Table 1 - AA: 10 parts by weight — Example 3 — Table 1 - SiMac: 10 parts by weight — Example 3 — Table 1 - Initiator-1: 0.004 parts by weight — Example 3 — Table 1 - IOTG: 0.08 parts by weight — Example 3 — Table 1 - ABP: 0.3125 parts by weight — Example 3 — Table 1 - IOA: 80 parts by weight — Example 4 — Table 1 - AA: 10 parts by weight — Example 4 — Table 1 - SiMac: 10 parts by weight — Example 4 — Table 1 - Initiator-1: 0.004 parts by weight — Example 4 — Table 1 - IOTG: 0.15 parts by weight — Example 4 — Table 1 - ABP: 0.3125 parts by weight — Example 4 — Table 1 - Additive mixture IOTG: 4.2 parts — Example 1 — Table 2 - Additive mixture Initiator-1: 0.5 parts — Example 1 — Table 2 - Additive mixture Initiator-2: 0.1 parts — Example 1 — Table 2 - Additive mixture Initiator-4: 0.15 parts — Example 1 — Table 2 - Additive mixture EA: 45.05 parts — Example 1 — Table 2 - Additive

mixture IOTG: 5.58 parts — Example 2 — Table 2 - Additive mixture Initiator-1: 0.5 parts — Example 2 — Table 2 - Additive mixture Initiator-2: 0.1 parts — Example 2 — Table 2 - Additive mixture Initiator-4: 0.15 parts — Example 2 — Table 2 - Additive mixture EA: 43.67 parts — Example 2 — Table 2 - Additive mixture IOTG: 7.44 parts — Example 3 — Table 2 - Additive mixture Initiator-1: 0.5 parts — Example 3 — Table 2 - Additive mixture Initiator-2: 0.1 parts — Example 3 — Table 2 - Additive mixture Initiator-4: 0.15 parts — Example 3 — Table 2 - Additive mixture EA: 41.81 parts — Example 3 — Table 2 - Additive mixture IOTG: 13.95 parts — Example 4 — Table 2 - Additive mixture Initiator-1: 0.5 parts — Example 4 — Table 2 - Additive mixture Initiator-2: 0.1 parts — Example 4 — Table 2 - Additive mixture Initiator-4: 0.15 parts — Example 4 — Table 2 - Additive mixture EA: 35.3 parts — Example 4 — Table 2 - First-stage heating temperature: 60°C — Examples 1-4 - First-stage peak temperature: 150°C — Examples 1-4 - Second-stage heating temperature: 60°C — Examples 1-4 - Second-stage peak temperature: 160°C — Examples 1-4 - Inherent viscosity: 0.55 dl/g — Example 1 — Table 3 - UV cure: 1 pass — Example 1 — Table 3 - Initial 180° peel adhesion: 8.6 N/dm — Example 1 — Table 3 - Adhesion after 24 hour dwell: 110.7 N/dm — Example 1 — Table 3 - Inherent viscosity: 0.62 dl/g — Example 2 — Table 3 - UV cure: 1 pass — Example 2 — Table 3 - Initial 180° peel adhesion: 1.9 N/dm — Example 2 — Table 3 - Adhesion after 24 hour dwell: 100.3 N/dm — Example 2 — Table 3 - Inherent viscosity: 0.37 dl/g — Example 3 — Table 3 - UV cure: 3 passes — Example 3 — Table 3 - Initial 180° peel adhesion: 40.3 N/dm — Example 3 — Table 3 - Adhesion after 24 hour dwell: 111.0 N/dm — Example 3 — Table 3 - Inherent viscosity: 0.29 dl/g — Example 4 — Table 3 - UV cure: 4 passes — Example 4 — Table 3 - Initial 180° peel adhesion: 8.6 N/dm — Example 4 — Table 3 - Adhesion after 24 hour dwell: 100.4 N/dm — Example 4 — Table 3 - Extrusion temperatures: 110, 115, 120, 140, 140°C — Example 10 - Extrusion pressure: 2,960 to 3,450 kPa — Example 10 - Extrusion speed: 90 RPM — Example 10 - Release value: 2.62 N/dm — Example 10 - Example 11 first-stage charge: MA 49 g, MMA 20 g, MAA 5 g, IOTG 0.3 g, MFS 25 g — Example 11 - Antioxidant-1: 0.1 g — Example 11 - Initiator-1: 0.1 g — Example 11 - Initiator-3: 0.1 g — Example 11 - First-stage heating temperature: 59°C — Example 11 - First-stage peak temperature: 170°C — Example 11 - Second-stage charge: Initiator-1 0.2 g, Initiator-2 0.2 g, Initiator-4 0.04 g, Initiator-6 0.2 g — Example 11 - Second-stage solvent: EA 19.36 g — Example 11 - Second-stage heating temperature: 59°C — Example 11 - Second-stage peak temperature: 190°C — Example 11 - Inherent viscosity: 0.25 dl/g — Example 11 - Grinder speed: 3000 rpm — Example 12 - Grinder sieve: 2.0 mm — Example 12 - Pellet drying: 65.5°C for 24 hours — Example 12 - Extrusion temperatures: 160, 170, 190°C — Example 12 - Layer ratio: approximately 1:2 top layer:bottom layer — Example 12 - Aging: 65°C for 10 days — Example 12 - Example 13 layer ratio: approximately 1:2 top layer:bottom layer — Example 13 - Aging: 65°C for 10 days — Example 13 - 180° peel after 10 days at 65°C: 32.2 N/dm — Example 12 - Readhesion after 10 days at 65°C: 22.5 N/dm — Example 12 - 180° peel after 10 days at 65°C: 34.0 N/dm — Example 13 - Readhesion after 10 days at 65°C: 16.4 N/dm — Example 13

Relevant Experimental Evidence - Example 1: a two-step adiabatic polymerization gave a polymer with IV 0.55 dl/g and strong post-cure adhesion. - Example 2: higher SiMac loading and additive composition changed the polymer response while maintaining high aged adhesion. - Example 3: increased additive mixture altered inherent viscosity to 0.37 dl/g and required 3 UV passes. - Example 4: the lowest IV in the set, 0.29 dl/g, was obtained with the highest additive

mixture loading. - Example 11: a second monomer set was polymerized adiabatically to IV 0.25 dl/g, showing broad applicability of the staged process.

Technical Relevance This patent is relevant as a process-control analogue: it explicitly uses chain transfer and staged adiabatic polymerization to tune polymer molecular weight and handling properties. While not NBR, the same control logic is directly applicable to low-MW NBR synthesis by radical solution or bulk routes.

Patent 4

- Patent Number: US10161556B2
- Patent Title: High temperature insulated pipelines
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose NBR synthesis; it discloses an insulated subsea transport conduit in which the first thermal insulation layer may comprise fluoropolymers, HNBR, or blends of fluoropolymers with HNBR. The focus is on selecting heat-resistant polymer compositions and crosslinking strategies for long-term high-temperature service. - Thermal conductivity: less than about 0.40 W/mK — Abstract/Claims - Operating temperature range: about 150°C to above about 205°C — Abstract/Claims - Thermal conductivity: about 0.15 to about 0.30 W/mK — Claims - First thermal insulation layer: solid — Claims - First thermal insulation layer: thermoset fluoroelastomer composition — Claims - Crosslinked propylene-tetrafluoroethylene copolymer — Claims - Fluoroelastomer molecular weight: about 100,000 to about 250,000 — Detailed description - Second thermal insulation layer: HNBR — Detailed description - Second thermal insulation layer: elastomer selected from HNBR, NBR, silicone rubber, EPDM rubber, and butyl rubber — Detailed description - Pipe insulation heat transfer coefficient range: 3-5 W/m²K — Background

Relevant Experimental Evidence - No numbered examples were detected by the parser. - The disclosure emphasizes HNBR as a high-temperature insulation material rather than as a synthesis target. - The patent frames thermal conductivity and heat resistance as the key design constraints for the polymer layer.

Technical Relevance This patent is only indirectly relevant to low-MW NBR because it identifies HNBR/NBR as elastomeric materials used in demanding thermal environments. It does not provide a route to low molecular weight NBR, but it helps define performance contexts where low-viscosity or modified nitrile elastomers may be useful.

Patent 5

- Patent Number: US12146207B2
- Patent Title: Degradable and/or deformable diverters and seals
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses degradable and/or deformable diverters and seals for wellbore applications, including embodiments where the core structure is formed of a rubber material such as nitrile rubber. The rubber component is used as part of a temporary sealing architecture with degradable or dissolvable behavior, not as a polymer synthesis target. - Polyvinyl butyral coating solution: 25 g PVB in 500 mL ethanol (5% w/v) — Example 1 - Dialdehyde (glyoxal): 0.1 wt.% — Example 1 - Coating thickness: 33 μm — Example 1 - Storage stability: over one year at 38°C — Example 1 - HCl slurry: 1.5 mM HCl, 100 mL — Example 1 - Heating: 100°C for 72 hours — Example 1 - Unconstrained adhesion strength: 104 psig — Example 1 - NBR particles: 20 vol.% — Example 1 - Composite ball diameter: ■ in. — Example 1 - Composite ball density: 0.95 g/cc — Example 1 - Pressure hold: 1500 psi for 2 hours — Example 1 - Weight loss: 50% over 72 hours in tap water at 51.7°C — Example 1 - Polyvinyl butyral coating solution: 30 g PVB in 500 mL ethanol (6% w/v) — Example 2 - Dialdehyde (glyoxal): 0.1 wt.% — Example 2 - Coating thickness: 39 μm — Example 2 - Storage stability: over one year at 38°C — Example 2 - HCl slurry: 1.5 mM HCl, 100 mL — Example 2 - Heating: 100°C for 72 hours — Example 2 - Unconstrained adhesion strength: 149 psig — Example 2 - NBR particles: 20 vol.% — Example 2 - Composite ball diameter: ■ in. — Example 2 - Composite ball density: 0.80 g/cc — Example 2 - Pressure hold: 1500 psi for 4 hours — Example 2 - Weight loss: 50% over 96 hours in tap water at 51.7°C — Example 2 - Polyvinyl butyral coating solution: 120 lb PVB in 180 gallons ethanol (10% w/v) — Example 3 - Dialdehyde (glyoxal): 0.1 wt.% — Example 3 - Coating thickness: about 49 μm — Example 3 - Storage stability: over a year at 38°C — Example 3 - HCl slurry: 1.5 mM HCl, 100 mL — Example 3 - Heating: 100°C for 72 hours — Example 3 - Unconstrained adhesion strength: 204 psig — Example 3 - Density of degradable rubber component: about 1.2 g/cc — Example 9 - Pressure hold: about 9 ksi — Example 9 - Degradation medium: 0.1% KCl water at 60°C — Example 9 - Degradation time: five days — Example 9 - Density of degradable rubber component: about 1.1 g/cc — Example 6 - Pressure hold: about 6 ksi — Example 6 - Degradation medium: di water at 60°C — Example 6 - Degradation time: five days — Example 6 - Pressure increase before sealing irregularities: between 1 and 2 ksi — Examples 6, 7, 9 - Pressure hold: about 8 ksi — Example 7 - Pressure increase before sealing irregularities: 500 to 800 psi — Example 8 - Pressure hold: about 3000 psi — Example 8

Relevant Experimental Evidence - Example 1: a PVB-coated particulate system and an NBR-containing composite ball were demonstrated, including 1500 psi pressure resistance. - Example 2: increased PVB loading and lower-density NBR composite gave longer weight-loss

time and higher adhesion strength. - Example 3: scale-up of the coating process produced a coated proppant with 204 psig unconstrained adhesion strength. - Example 7: an NBR rubber component sealed irregular holes and held about 8 ksi. - Example 8: an NBR component with a different geometry sealed irregular holes and held about 3000 psi.

Technical Relevance This patent is relevant because it explicitly uses NBR as a rubber component in deformable, temporary sealing devices. Although it does not address low molecular weight NBR synthesis, it shows an application space where low-viscosity or processable NBR grades could be valuable.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2921509B1 is the only patent in the set that explicitly targets liquid, low-Mn polymer products; its disclosed Mn range spans 2080 to 5385 in the examples, with Mw/Mn around 1.81-1.89 and low bromine number (<0.1 in the inventive examples).
- US10465017B2 uses staged essentially adiabatic polymerization and explicit chain-transfer/initiator management to tune polymer inherent viscosity from 0.25 to 0.62 dl/g, illustrating a molecular-weight-control strategy that is conceptually transferable to low-MW NBR.
- EP2389295B1 is formulation-driven rather than synthesis-driven: it reports Mooney viscosity values of 55.6-80.0 ME and mixing at 70-154°C, showing how diene-rubber processability is adjusted downstream of polymerization.
- US10161556B2 and US12146207B2 are application patents rather than synthesis patents; both mention NBR/HNBR as functional elastomeric materials, but neither provides a route to low molecular weight NBR.
- Across the set, solution polymerization appears only in EP2921509B1 and in the background discussion of EP2389295B1; the other patents rely on bulk/adiabatic polymerization or end-use compounding.
- No patent in the manifest discloses a dedicated NBR chain-transfer recipe, acrylonitrile/butadiene feed ratio, or NBR-specific molecular-weight target; the closest transferable controls are catalyst/initiator loading, hydrogen or chain-transfer use, and temperature management.

4. CONCLUSION

The retrieved patent set does not contain a direct low molecular weight NBR synthesis disclosure, but it does identify the most relevant control concepts for achieving such a material: solution polymerization, staged radical or catalytic initiation, chain-transfer management, and tight thermal control. EP2921509B1 provides the strongest low-MW process analogue through liquid copolymer production with Mn in the low-thousands and narrow dispersity, while US10465017B2

demonstrates how staged adiabatic polymerization and chain-transfer agents can be used to tune polymer viscosity. For low molecular weight NBR development, the evidence supports targeting solution polymerization with deliberate molecular-weight suppression, low-to-moderate temperatures, and explicit chain-transfer/initiator control, followed by formulation or compounding only after the desired low-viscosity backbone is obtained.

5. REFERENCES

- EP2389295B1 | Rubber mixtures containing silane and having possibly functionalized diene ... | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2389295B1/en>
- EP2921509B1 | Method for preparing ethylene and alpha-olefin copolymers | Not disclosed | EP | 2025 | <https://patents.google.com/patent/EP2921509B1/en>
- US10465017B2 | Bulk polymerization of silicone-containing copolymers | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10465017B2/en>
- US10161556B2 | High temperature insulated pipelines | Not disclosed | US | 2018 | <https://patents.google.com/patent/US10161556B2/en>
- US12146207B2 | Degradable and/or deformable diverters and seals | Not disclosed | US | 2024 | <https://patents.google.com/patent/US12146207B2/en>